1 Solution — GIAM 5.2 Problem 5

1.1 Problem

Prove the following formula for a product:

$$\prod_{i=2}^{n}\left(1-\frac{1}{i}\right)=\frac{1}{n}, \qquad n \geq 2.$$

1.2 Direct Proof (Telescoping)

Rewrite each factor as a single fraction:

$$1-\frac{1}{i} \;=\; \frac{i-1}{i}.$$

Therefore

$$\prod_{i=2}^{n}\left(1-\frac{1}{i}\right) \;=\; \prod_{i=2}^{n}\frac{i-1}{i} \;=\; \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdots\frac{n-1}{n}.$$

Every numerator after the first cancels the denominator of the previous factor (a telescoping product), leaving

$$\prod_{i=2}^{n}\left(1-\frac{1}{i}\right)=\frac{1}{n},$$

as claimed.

1.3 Proof by Induction

Basis $(n=2)$:

$$\prod_{i=2}^{2}\left(1-\frac{1}{i}\right)=1-\frac{1}{2}=\frac{1}{2}=\frac{1}{n}.$$

Inductive step. Assume, for some integer $k \geq 2$,

$$\prod_{i=2}^{k} \left(1 - \frac{1}{i}\right) = \frac{1}{k}.$$

Then

$$\prod_{i=2}^{k+1} \left(1 - \frac{1}{i}\right) = \left[\prod_{i=2}^{k} \left(1 - \frac{1}{i}\right)\right] \cdot \left(1 - \frac{1}{k+1}\right) = \frac{1}{k} \cdot \frac{k}{k+1} = \frac{1}{k+1}.$$

By the principle of mathematical induction, the identity holds for all integers $n \geq 2$.